

IRFAN SHARIF

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SUMMARY

Built GPU clusters, admission control systems, storage/replication engines; published in SIGMOD

EDUCATION

University of Waterloo

Sept '14 – Apr '19

Bachelor of Applied Science in Honours Computer Engineering (with Distinction)

EXPERIENCE

Arcol

Toronto, ON

Member of Technical Staff

Jan '26 – Present

- Collaborative architectural design, through graphics and geometry

Modal Labs

New York, NY

Member of Technical Staff

Sept '23 – Nov '25

- Built algorithmic control systems for a 20,000 GPU cluster, managing millions in annual spend
- Designed a global container scheduler with intelligent routing/affinities, and sub-second placement
- Achieved <30s gang scheduling for training clusters, through novel de-fragmentation primitives

Cockroach Labs

Toronto, ON

Senior Software Engineer (Transactional KV)

July '19 – Sept '23

- Led admission control: CRDB's overload protection, work prioritization, and fairness mechanism
- Built an elastic CPU limiter patching Go, protecting latencies from in-process background work
- Improved schema count scalability from $O(10k)$ to $O(1M)$, re-architecting for multi-tenancy
- Worked on flow control for quorum writes, upgrades infrastructure, bazel-ifying monorepo, tracing

LinkedIn

Sunnyvale, CA

Software Engineering Intern (Machine Learning Infrastructure)

Sept – Dec '18

- Built systems for model serving, versioning, runtime and deployment across thousands of hosts

Amazon

Palo Alto, CA

Software Engineering Intern (Search Relevance)

Jan – Apr '18

- Used sequence-to-sequence learning for query reformulation, improving recall and ranking models

Cockroach Labs

New York, NY

Backend Engineering Intern (Storage, Distributed SQL)

Aug '16 – Sept '17

- Designed a dedicated storage engine for Raft, contributed to TLA+ verified PreVote extension
- Implemented aggregations, sorting, n -way parallel joins, and row de-duplications for query engine

Shopify

Ottawa, ON

Production Engineering Intern (Infrastructure)

Jan – Apr '16

PUBLICATIONS

- [1] SIGMOD 2020. *CockroachDB: The Resilient Geo-Distributed SQL Database*
- [2] SIGMOD 2022. *Enabling the Next Generation of Multi-Region Applications with CockroachDB*
- [3] SIGMOD 2025. *CockroachDB Serverless: Sub-second Scaling with Cluster Virtualization*
- [4] US Patent 2025. *Admission Control for Multi-Consensus-Based Replication*
- [5] Hydra Conference 2021. *Hitchhiker's Guide to Transactions*

RESEARCH

Waterloo AI Institute – Studied SAT solvers, with Prof. Vijay Ganesh

Oct – Dec '16

PROJECTS

- **Kura** – An E2EE, synchronized, distributed file system with live-sharing and file streaming